

# Detailed Project Report (DPR)

## Coconut Cider Vinegar — Food-Grade Fermentation Unit from Recovered Coconut Water (starting ~2,000 nuts/day, ramping to ~6,000 nuts/day in place)

**Promoter:** Arun T. (sole proprietary concern) **Location:** Eruthenpathy, Chittur Taluk, Palakkad District, Kerala (Kerala-Tamil Nadu border) **Raw material:** Coconut water recovered at the village aggregation point **before the husked nuts are sent to Kangeyam (Tamil Nadu) for drying/copra** — a stream that is currently discarded. **Structure:** A **single registered food-processing unit** that **starts small (~2,000 nuts/day of coconut water) and ramps in place to ~6,000 nuts/day**, housed in a **purpose-built ~8,000 sq ft food-grade hall** on the promoter's own land at the coconut collection yard. Designed for the **lenient pollution category with Zero-Liquid-Discharge**, and sized to capture the **full ₹40 lakh Kerala ESS capital subsidy** on top of central interest-subvention and credit-guarantee support. **Prepared for:** Submission to District Industries Centre (DIC), Palakkad; FSSAI; Kerala State Pollution Control Board (KSPCB); and the lending bank (AIF / term loan). **Document status:** Planning-grade DPR — figures are indicative estimates to be firmed up with (a) a coconut-water throughput and lab analysis, (b) formal machinery quotations, (c) written KSPCB categorisation confirmation, and (d) written DIC/bank confirmation of ESS + AIF convergence, before sanction.

### 1. Executive summary

This project establishes a **coconut cider vinegar** business as a **sole proprietary concern of Arun T.**, converting the **coconut water that is presently thrown away** into a shelf-stable, value-added food product.

- **The core insight — recover a wasted stream at zero raw-material cost.** The village already de-husks coconuts and ships them to **Kangeyam** for sun-drying into copra (local drying is impossible for ~6 months of the year due to rain). At Kangeyam the nuts are split and **the water is drained onto the ground and lost**. By **draining the water at the village collection yard first** and only then dispatching the nuts to Kangeyam, the project captures ~0.3 litre of free raw material per nut with **no loss to the copra business**.
- **Climate fit — the monsoon that blocks drying does NOT block vinegar.** Fermentation happens in **closed, temperature-controlled tanks indoors**, so the ~6-month rainy season that forces the village to outsource drying is irrelevant to this unit. The plant runs year-round in the same climate that rules out sun-dependent products.
- **Constitution:** sole proprietorship — **Arun T., proprietor**.
- **One unit, phased in place (not split):** start at **~2,000 nuts/day (~600 L water/day)** and **ramp to ~4,000 then ~6,000 nuts/day** by adding modular fermentation/acetification capacity within the **same registered premises**.
- **Purpose-built hall:** a **~8,000 sq ft food-grade hall** is built **once, up front, for the ~6,000 nuts/day endgame** (hygienic epoxy floor, drained wet-process zone, separate clean bottling zone, warehouse, QC lab). Sizing the civil + utilities + ETP for the endgame puts fixed capital just past the point that **captures the full ₹40 lakh Kerala ESS subsidy** (the scheme maximum) without overspending.
- **Pollution category — designed for the lenient regime with Zero-Liquid-Discharge (ZLD).** Effluent is dilute, biodegradable wash/CIP water; it is treated in a compact ETP and **reused on the greenbelt/plantation — nothing is discharged to land or water**. Solid residues (filter cake, spent yeast/mother) are **composted / diverted to the promoter's biochar unit**. A **written KSPCB categorisation confirmation** is sought (Section 7).
- **Product: coconut cider vinegar** in two lines — (a) **raw, unfiltered "with the mother"** (premium, apple-cider-vinegar-style health positioning) and (b) **filtered food-grade vinegar** in bulk for the pickle / food-processing / HoReCa trade (the volume anchor).
- **Regulatory backbone: FSSAI "brewed vinegar" standard** — minimum **3.75% acetic acid** (target 5%), no synthetic-acid fortification (Section 6, 10).
- **Project cost (Phase-1): ~₹1.01 crore** (~₹0.93 cr fixed capital + ₹8 L working-capital margin); land is the promoter's own (in-kind, excluded).
- **Financing & maximised subsidy stack:** promoter equity + **AIF term loan (3% interest subvention, 7 yrs) + CGTMSE collateral-free cover + Kerala ESS capital subsidy (~₹40 L — the scheme maximum) + PMFME/ODOP branding & marketing grant on the separate marketing head** (Section 12).
- **Cluster synergy:** shares land, utilities, QC and management with the promoter's **biochar / bio-fertilizer unit**; coconut **husk and shell** feed the biochar unit and vinegar **process residues** feed its compost/enrichment line (Section 16).
- **Indicative payback:** ~2-3 years on the bulk (food-industry) base case; faster as the branded "with-mother" line and exports scale.

### 2. Promoter

**Promoter / constitution:** Arun T. — sole proprietary concern (proprietorship).

- Owns the project land at Eruthenpathy, Chittur (the coconut collection yard used for aggregating husked nuts before dispatch to Kangeyam) and **~10 acres** of coconut holding nearby — giving both a **captive raw-material base** and space for the unit, greenbelt and ETP reuse.
- Contact: 4/296, Mangapallam South, RVP Pudur Post, Palakkad District, Kerala – 678555 · 9901955667 · er.arunt@gmail.com.
- Already developing a **biochar / enriched bio-fertilizer unit** on the adjacent 38-cent plot — this vinegar unit extends the same **coconut value-addition cluster** and shares overheads.

### 3. Project concept & rationale

Every mature coconut sent for copra carries ~0.3 litre of coconut water. In the present flow — **de-husk → ship whole nuts to Kangeyam → split & dry** — that water is **drained and wasted at Kangeyam**. Recovering it at source and fermenting it into vinegar:

1. **Monetises a zero-cost waste stream** without touching the existing copra income (the kernel still goes to Kangeyam exactly as before).
2. Produces a **shelf-stable, high-margin food product** (vinegar keeps for years — no cold chain), unlike the perishable water itself.
3. **Fits the local climate**: enclosed-tank fermentation is unaffected by the ~6-month monsoon that makes local sun-drying impossible.
4. **Qualifies for strong central + state subsidy** as an agro-based micro/small food-processing MSME (Section 12).
5. Is structured for the **lenient pollution category with ZLD**, so the unit can be established and grown in place under a light-touch consent regime (Section 7).

#### 3.1 The "drain-first" logistics change (answering the core question)

Instead of shipping whole nuts, the yard will **drain the water and then dispatch the nuts to Kangeyam**. Two draining methods, with a clear recommendation:

| Method                               | How                                                              | Trade-off                                                                                                                                                              |
|--------------------------------------|------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Full split                           | Halve the nut, drain, ship halves                                | Maximum water recovery, but the <b>exposed kernel is vulnerable to mould/rancidity in humid transit</b> — a real copra-quality (and aflatoxin) risk during the monsoon |
| <b>Bored-eye drain (recommended)</b> | Pierce the soft "eye", drain the water, ship the near-intact nut | Recovers the water while the <b>shell keeps protecting the kernel</b> to Kangeyam — protects copra grade                                                               |

**Recommendation: bore-and-drain, keeping the shell intact**, so copra quality is preserved. Advantages captured: (a) free vinegar raw material; (b) **lower transport weight to Kangeyam** (~0.3 kg/nut of water removed); (c) less uncontrolled in-transit fermentation/leakage; (d) the copra process is otherwise unchanged. Confirm the commercial arrangement with the Kangeyam driers (whether they pay per nut or per kg — draining changes the weight).

### 4. Market analysis & best markets

Coconut vinegar is a traditional product (the Philippines is the world hub) with a modern **health-food tailwind** — it is positioned like **apple cider vinegar (ACV)**. Two segments, sold best-first:

| Segment                                                                                           | Indicative ex-factory price                        | Role                                                               |
|---------------------------------------------------------------------------------------------------|----------------------------------------------------|--------------------------------------------------------------------|
| <b>Bulk filtered food-grade vinegar</b> (drums/cans) to pickle & food-processing industry, HoReCa | <b>₹40-70 /L</b>                                   | <b>Volume anchor</b> — easy to sell, absorbs output from day one   |
| <b>Branded raw "coconut cider vinegar, with the mother"</b> (glass bottles, 250/500 ml)           | <b>₹120-160 /L wholesale</b> (retails ₹200-400 /L) | <b>Margin driver</b> — health/wellness, organic stores, e-commerce |
| <b>Export</b> (Middle East, Europe, US health market)                                             | Premium                                            | Upside, needs APEDA + organic certification                        |

#### Best markets, best-first:

1. **Domestic bulk to the pickle & food-processing trade (primary volume)**. Tamil Nadu/Kerala have a large pickle, sauce and food-service industry that buys vinegar in bulk. Lowest marketing cost; anchors capacity utilisation.
2. **Branded ACV-style "with the mother" (primary margin)**. Ride the wellness trend; sell through organic stores, pharmacies, and **e-commerce (Amazon/Flipkart)**. Requires branding + FSSAI-compliant labelling.
3. **ODOP / institutional channels**. Position under **PMFME One-District-One-Product** branding support; supply to Kerala/TN government emporia and FPO market linkages.
4. **Export (upside)**. Coconut vinegar has niche export demand; pursue via **APEDA** registration and **organic (NPOP/Jaivik Bharat)** certification once volumes and consistency are proven.

**Demand strategy & honest caveat:** production is the easy part — **market development and distribution are the real constraint** for a niche product. The plan therefore **anchors on bulk food-industry sales** (absorbable, low-marketing) and treats the branded consumer line and exports as the margin/growth upside, funded partly by the **branding & marketing subsidy** (Section 12).

## 5. Raw-material plan (coconut water)

| Source                                                      | Basis                                                                                                                                     |
|-------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Own ~10-acre coconut holding</b>                         | Captive / free — nuts routed through the yard for de-husking + water draining before Kangeyam dispatch                                    |
| <b>Village cluster nuts aggregated for Kangeyam</b>         | The yard already collects husked nuts for dispatch; water is drained here at near-zero cost under a supply understanding with the growers |
| <b>Jaggery / muscovado sugar</b> (process input, not waste) | Open-market — needed to raise the fermentable sugar level (see below)                                                                     |

**Perishability — the binding operational rule.** Coconut water spoils within hours. It must be **filtered and pasteurised the same day it is drained**. Throughput is therefore matched to the daily nut flow through the yard, not stockpiled.

**Composition & why sugar must be added.** Mature (copra-grade) coconut water is thin — about **330 mL/nut at ~4.85 °Brix (~2.5 g sugar/100 mL)** — too low to ferment directly to legal-strength vinegar. It is **adjusted with jaggery/muscovado to ~15 °Brix** before fermentation, to reach ~7–8% alcohol and then  $\geq 5\%$  acetic acid. This is standard practice; note the FSSAI rule that the acidity must come from **fermentation, not from added synthetic acetic acid** (Section 6).

Execute a **written raw-material understanding** with the growers/aggregators routing nuts to Kangeyam, so water recovery is secured at source.

## 6. Technical design, process & capacity (single unit, phased in place)

### 6.1 Ramp path (one unit, capacity added in place — not split)

| Phase                    | Coconut water intake | Nuts/day equiv. | Vinegar output | Annual output (300 days) |
|--------------------------|----------------------|-----------------|----------------|--------------------------|
| <b>Phase-1 (start)</b>   | ~600 L/day           | ~2,000          | ~600 L/day     | ~1.8 lakh L/yr           |
| <b>Phase-2</b>           | ~1,200 L/day         | ~4,000          | ~1,200 L/day   | ~3.6 lakh L/yr           |
| <b>Phase-3 (endgame)</b> | ~1,800 L/day         | ~6,000          | ~1,800 L/day   | ~5.4 lakh L/yr           |

Capacity is grown by **adding modular fermenters/acetators inside the same hall** — the unit is never split into separate registrations. The **hall, utilities, boiler, ETP and bottling line are built once for the endgame**, so ramping needs only extra tankage.

### 6.2 Process (two-stage fermentation)

```
Drained coconut water (same day)
-> Filter / screen -> Pasteurise 60-65 C, 10-15 min -> cool
-> Adjust to ~15 Brix with jaggery/muscovado; add yeast nutrient
-> STAGE 1 Alcoholic fermentation (Saccharomyces cerevisiae),
    anaerobic, ~7-14 days -> "coconut cider" ~7-8% ABV
-> Rack / decant off yeast; pasteurise; cool
-> STAGE 2 Acetous fermentation (Acetobacter aceti + mother of vinegar),
    aerobic; slow surface method ~3-4 wk OR submerged acetator ~1-2 wk
    -> vinegar, acidity  $\geq 5\%$ 
-> Age / clarify -> filter -> standardise acidity
-> LINE A (raw): keep unfiltered "with the mother", cold-fill (premium)
    LINE B (food-grade): pasteurise, filter bright, bulk-fill
-> Bottle / label / batch-code -> QC release -> warehouse
```

### 6.3 Product quality & FSSAI compliance

To be sold legally as vinegar in India (FSSAI "brewed vinegar" standard), each batch must meet: - **Acidity  $\geq 3.75\%$  (w/v) as acetic acid** (the plan targets ~5% for a robust product), - **Total solids  $\geq 1.5\%$ , total ash  $\geq 0.18\%$** , - **No mineral acids**, and it **must not be fortified with synthetic acetic acid** — the acid must come from fermentation.

**QC lab** performs titratable-acidity titration, pH, residual-alcohol and basic microbiological checks on every batch, with batch coding for traceability.

### 6.4 Climate & utility note

Fermentation is enclosed and mildly temperature-controlled; the monsoon is a non-issue. Main utilities: potable/softened water, a small steam boiler for pasteurisation/CIP, 3-phase power, and a compact cold room for the raw "with-mother" line.

## 7. Pollution-control category & consent strategy

**Objective:** establish the unit in the lenient pollution category — legitimately, on the merits — with Zero-Liquid-Discharge.

### 7.1 Pollution profile (genuinely low)

| Stream          | Nature                                                                                    | Management                                                                                                                                                                  |
|-----------------|-------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Liquid effluent | Dilute, biodegradable <b>wash / CIP / floor-wash water</b> ; small volume                 | Compact <b>ETP</b> (screening → equalisation → anaerobic + aerobic → clarifier) → <b>treated water reused on greenbelt/plantation. ZLD — no discharge to land or water.</b> |
| Solid residue   | Filter cake, spent yeast, surplus "mother"                                                | <b>Composted / diverted to the biochar-bio-fertilizer unit</b> (Section 16)                                                                                                 |
| Air             | Minor steam / faint fermentation odour; no combustion process gases beyond a small boiler | Enclosed process; boiler stack to norms; greenbelt buffer                                                                                                                   |
| Hazard          | None significant — no solvents, no synthetic acids, no alcohol bottled for sale           | —                                                                                                                                                                           |

### 7.2 Categorisation strategy

Food fermentation can be viewed two ways by the Board, so this is handled explicitly rather than assumed: - The **RED "fermentation industry"** entry in the CPCB/KSPCB list targets **distilleries, breweries and malt units producing potable/industrial alcohol**. This unit is **not** that: **no alcohol is bottled or sold** (ethanol is only a fleeting in-process intermediate), scale is small, and effluent is minor and organic. - General **food & food-processing** activity of this size, with **ZLD and a working ETP**, fits the **Green/Orange (lenient) regime**. - **Action:** design to the lenient category on the merits (ZLD, ETP, no discharge) and **obtain a written KSPCB categorisation confirmation** *before* sanction — mirroring the approach used for the promoter's biochar unit. Sanction should follow that written confirmation.

### 7.3 Consent path

- **CTE (Consent to Establish) then CTO (Consent to Operate)** from KSPCB under the Water Act 1974 and Air Act 1981; amend the CTO upward on each ramp.
- **No Environmental Clearance** (not an EIA-2006 scheduled activity at this scale).
- Operating without CTE/CTO is an offence under the EPA 1986 — consents are obtained before commissioning.

## 8. Machinery & suppliers (Phase-1, expandable)

Food-grade **SS-304** contact equipment throughout. Preferred sourcing from the **Coimbatore-Pollachi-Palakad hub** (30-60 km) for service and spares.

| Block                   | Key equipment                                                                                                                    |
|-------------------------|----------------------------------------------------------------------------------------------------------------------------------|
| Water recovery          | Nut de-husk/bore-drain station, collection tanks, plate filter, chiller                                                          |
| Thermal                 | Steam boiler (small), plate pasteuriser / heat exchanger                                                                         |
| Fermentation            | SS jacketed <b>alcoholic fermenters; acetators / surface-acetification vats</b> with mother-of-vinegar culture; airlift/aeration |
| Finishing               | Clarifier, cartridge/plate filter, acidity-standardising tank                                                                    |
| Packing                 | Bottle rinser, filler, capper, labeller, batch coder; drum/can bulk-fill                                                         |
| Utilities & environment | Water softener/RO, CIP set, cold room, <b>ETP + ZLD reuse</b> , weighing, fire                                                   |
| Lab                     | Titration set, pH meter, alcoholometer, incubator                                                                                |

**Recommendation:** a compact Phase-1 line (~600 L/day) with **pre-provisioned tie-ins for Phase-2/3 tankage**, so the unit grows in place. Fillable vendor comparison in **Annexure D**.

## 9. Land, layout & civil works

The unit sits on the promoter's own yard/land, laid out for **FSSAI-grade hygienic flow** (raw/"dirty" wet zone kept separate from the clean bottling zone to prevent cross-contamination):

- **Wet-process zone** (water intake, pasteurisation, fermentation) — drained, epoxy floor, sloped to ETP.

- **Clean zone** (clarification, bottling, labelling, warehouse) — positive-flow, restricted access.
- **QC lab + office** — fixed rooms; **personnel hygiene** (change/handwash) at the entry to the clean zone.
- **Utilities yard** — boiler, softener, ETP with **treated-water reuse to the greenbelt/plantation**.

Building the ~8,000 sq ft hall once (for the endgame) avoids re-construction on ramp and lifts fixed capital past the ESS-cap point. Balance of plot is greenbelt (ETP reuse), parking and setbacks. **Pre-checks:** KSPCB setbacks; road access for tankers/dispatch; 3-phase power; water; land-use where applicable.

## 10. Regulatory & statutory compliance

| Approval                                                   | Authority                         | Notes                                                                        |
|------------------------------------------------------------|-----------------------------------|------------------------------------------------------------------------------|
| Udyam (MSME) registration                                  | Online                            | Gateway for all schemes                                                      |
| <b>FSSAI licence</b>                                       | FSSAI (State/Central by turnover) | Core food-safety licence; label approval                                     |
| <b>Consent to Establish / Operate (CTE/CTO)</b>            | KSPCB                             | Lenient category + ZLD; amend upward on ramp (Section 7)                     |
| <b>Legal Metrology — packaged commodity</b> registration   | Legal Metrology Dept.             | Mandatory declarations on bottles (net qty, MRP, batch, mfg date, FSSAI no.) |
| Factory / establishment registration                       | Labour / Factories Dept.          | As applicable by power & headcount                                           |
| Building permit + D&O (Dangerous & Offensive) licence      | Grama Panchayat                   | Hall / trade licence                                                         |
| Fire NOC                                                   | Fire & Rescue Dept.               | Boiler / covered hall                                                        |
| Boiler registration                                        | Dept. of Boilers                  | If steam boiler above threshold                                              |
| Groundwater NOC                                            | CGWA / State                      | If borewell in a notified/regulated area                                     |
| GST registration                                           | GST                               |                                                                              |
| APEDA registration + <b>Organic (NPOP / Jaivik Bharat)</b> | APEDA / accredited body           | <b>Only if/when exporting or making an organic claim</b>                     |

## 11. Cost of project (Phase-1)

| #  | Item                                                                                                                                  | ₹ Lakh              |
|----|---------------------------------------------------------------------------------------------------------------------------------------|---------------------|
| 1  | Food-grade hall ~8,000 sq ft (hygienic epoxy floor, drained wet zone, clean bottling zone, warehouse, QC lab — built for the endgame) | 42                  |
| 2  | Fermentation & acetification — SS fermenters + acetators (Phase-1, expandable)                                                        | 14                  |
| 3  | Coconut-water recovery — bore/drain station, collection tanks, filtration, chiller                                                    | 6                   |
| 4  | Boiler + pasteuriser / heat exchangers                                                                                                | 5                   |
| 5  | Clarification, filtration, bottling & labelling line, bulk-fill                                                                       | 8                   |
| 6  | <b>ETP + Zero-Liquid-Discharge reuse system</b>                                                                                       | 5                   |
| 7  | Electrical, water (softener/RO), fire, cold room, weighing                                                                            | 5                   |
| 8  | QC lab + office                                                                                                                       | 2                   |
| 9  | Site development (compound wall, gates, paving, greenbelt)                                                                            | 4                   |
| 10 | Preliminary & pre-operative (FSSAI, CTE/CTO, DPR, cultures, trials)                                                                   | 2                   |
|    | <b>Fixed capital sub-total</b>                                                                                                        | <b>93</b>           |
| 11 | Margin money for working capital                                                                                                      | 8                   |
|    | <b>Total project cost (Phase-1)</b>                                                                                                   | <b>~₹1.01 crore</b> |

**Land** is the promoter's own (in-kind) and is **excluded**. **Ramp add-ons (from cash flows / top-up AIF loan):** Phase-2 ≈ + ₹12 L and Phase-3 ≈ + ₹12 L (extra tankage only — hall, boiler, ETP and bottling line are already built).

## 12. Means of finance & maximised subsidy stack (Phase-1)

| Source                                                                                     | ₹ Lakh     | %           |
|--------------------------------------------------------------------------------------------|------------|-------------|
| Promoter equity (cash)                                                                     | 27         | 27%         |
| Term loan — bank, under <b>AIF (3% interest subvention, 7 yrs), CGTMSE</b> collateral-free | 66         | 65%         |
| Working capital limit (bank cash credit)                                                   | 8          | 8%          |
| <b>Total</b>                                                                               | <b>101</b> | <b>100%</b> |

## 12.1 How the subsidy is maximised

The plan is engineered so that **one large capital subsidy is captured at its ceiling, and only non-overlapping supports are stacked on top** (two capital subsidies cannot fund the same asset — "double-dipping" is disallowed and voids both).

- **Anchor = Kerala ESS capital subsidy — target ~₹40 L (the scheme maximum).** At the ~45% band (25% base for young/women/SC-ST/NRK entrepreneur + 10% agro-based priority sector + 10% new-technology/first-generation), the ₹40 L cap is reached at ~₹89 L eligible fixed capital. The project's ~₹93 L fixed capital just clears that point, so the full ₹40 L is captured without overspending. (*Palakkad does not attract the extra district bonus.*) ESS is received as **reimbursement after commissioning** and applied to **prepay the term loan**.
- **+ AIF (central) — 3% interest subvention (7 yrs) + CGTMSE guarantee.** This is an *interest/credit* benefit on a **different basis**, so it **stacks legitimately with ESS**. Effective term-loan rate  $\approx 9\% - 3\% = \sim 6\%$ ; the loan is **collateral-free** (land not mortgaged).
- **+ PMFME / ODOP branding & marketing grant — on the separate marketing head.** PMFME's branding-and-marketing support (up to 50% of eligible marketing spend, and ODOP linkage for coconut products) funds a **different expense (marketing, not plant)**, so it can be added without disturbing ESS on the capital.

## 12.2 Why ESS is the right anchor (decision logic)

| Scheme                                                      | Rate                      | Ceiling      | Subsidy on ~₹93 L project | Verdict                                            |
|-------------------------------------------------------------|---------------------------|--------------|---------------------------|----------------------------------------------------|
| <b>Kerala ESS</b>                                           | up to ~45%                | <b>₹40 L</b> | <b>~₹40 L</b>             | <b>Anchor — largest for this size</b>              |
| CDB Technology Mission on Coconut (product diversification) | ~25%                      | ₹50 L        | ~₹23 L                    | Beats ESS only if project > ~₹1.6 cr               |
| PMFME (micro food processing) capital subsidy               | 35%                       | ₹10 L        | ~₹10 L                    | Smaller cap; use its <i>marketing</i> head instead |
| PMKSY unit scheme                                           | 35% (50% difficult areas) | ₹5 cr        | (large-unit scheme)       | Consider only if scaling to a much larger plant    |

- **For this ~₹1 crore plant, Kerala ESS (₹40 L) is the maximum** capital subsidy. CDB or PMKSY would only win at much larger project sizes; if the promoter later scales past ~₹1.6 cr, re-evaluate CDB/PMKSY as the anchor.
- **Do NOT use PMEGP** as a capital subsidy — availing it disqualifies the unit from other government subsidies and would **void the far larger ESS + AIF benefit**.
- **Confirm ESS + AIF convergence in writing** with DIC Palakkad and the lending bank before sanction.

## 12.3 Net effect

ESS ₹40 L prepays the loan, cutting effective debt to ~₹26 L; AIF trims interest to ~6%; CGTMSE removes the collateral requirement; PMFME/ODOP part-funds branding. **Net capital at risk (Phase-1)  $\approx$  ₹61 L** after ESS.

## 13. Working capital (indicative, annual, Phase-1)

Jaggery/muscovado, yeast + Acetobacter cultures, bottles/caps/labels and bulk containers, wages, power/fuel, QC consumables, **marketing/distribution**, and ~1–2 months finished-goods holding. Phase-1 operating cost  $\approx$  **₹60–65 L/yr** (packaging + marketing are the largest heads; raw coconut water is ~free). Scales with capacity on ramp.

## 14. Profitability & ROI (indicative)

**Common assumptions:** 300 operating days; ~0.3 L water/nut; ~1:1 water-to-vinegar; jaggery ~₹4.5/L of product; effective interest 6%. Two cases by product mix.

| Scenario                                             | Output     | Blended price | Revenue  | Op. cost | Net (approx)     |
|------------------------------------------------------|------------|---------------|----------|----------|------------------|
| <b>Phase-1 — base (bulk-dominant)</b>                | 1.8 lakh L | ₹55/L         | ₹99 L    | ₹63 L    | <b>~₹36 L</b>    |
| <b>Phase-1 — upside (more branded "with mother")</b> | 1.8 lakh L | ₹80/L         | ₹1.44 cr | ₹70 L    | <b>~₹74 L</b>    |
| <b>Endgame (~6,000 nuts/day) — base</b>              | 5.4 lakh L | ₹55/L         | ₹2.97 cr | ₹1.7 cr  | <b>~₹1.27 cr</b> |



**Indicative payback**  $\approx$  2-3 years on the base (bulk) case, faster on the branded/export mix. Margins are structurally high because the **raw material is a recovered waste stream**; the swing factor is **market development**, which the branding subsidy part-funds. Treat the bulk case as the bankable base; branded/export is upside. **DSCR (Year 2) is comfortably bankable** (Annexure C).

## 15. Implementation schedule (sequence, not calendar)

1. Confirm daily nut throughput at the yard and coconut-water lab analysis  $\rightarrow$  fixes Phase-1 sizing.
2. Udyam registration; DIC/ESS pre-consultation; raw-material understanding with growers/aggregators.
3. **Written KSPCB categorisation confirmation** (lenient + ZLD); **FSSAI** application.
4. **CTE (KSPCB)**; hall design/build (endgame-sized, hygienic zoning); machinery order (Phase-1 line + ramp tie-ins).
5. Civil + utilities + ETP; Legal Metrology, fire NOC, boiler registration, Panchayat permits.
6. AIF term loan + CGTMSE; commissioning trials; **CTO**.
7. Culture establishment (yeast + mother of vinegar); first QC-released batches (acidity  $\geq 5\%$ ).
8. Bulk sales to pickle/food industry (base); launch branded "with-mother" line + e-commerce.
9. **ESS claim (max ₹40 L)** after commissioning  $\rightarrow$  prepay loan; PMFME/ODOP branding claim.
10. **Ramp in place** to Phase-2/3; pursue APEDA + organic certification for export.

## 16. Cluster synergy with the biochar / bio-fertilizer unit

The vinegar unit is not standalone — it completes a **coconut value-addition cluster** with the promoter's biochar unit, sharing land, utilities, QC lab and management:

- **Coconut husk & shell** removed at de-husking  $\rightarrow$  **feedstock for the biochar unit**.
- **Vinegar process residues** (filter cake, spent yeast/mother)  $\rightarrow$  **compost / enrichment feed** for the bio-fertilizer line, keeping the vinegar unit at **ZLD/zero-solid-waste**.
- **Shared overheads** (QC lab, weighbridge, office, power, management) lower unit costs for both.
- A single "**coconut zero-waste**" **brand story** (water  $\rightarrow$  vinegar, husk/shell  $\rightarrow$  biochar, kernel  $\rightarrow$  copra) strengthens marketing and any export/ESG positioning.

## 17. Value addition & product range

- **Coconut cider vinegar "with the mother"** — flagship health line (raw, unfiltered, cold-filled).
- **Filtered food-grade vinegar** — bulk for pickle/food industry, HoReCa.
- **Flavoured / infused** variants (ginger, garlic, chilli) for retail differentiation.
- **Optional future line — nata de coco** (edible bacterial-cellulose dessert gel, also made from coconut water) as a second high-value use of the same feedstock — evaluate after the vinegar line is stable.

## 18. Risks & mitigation

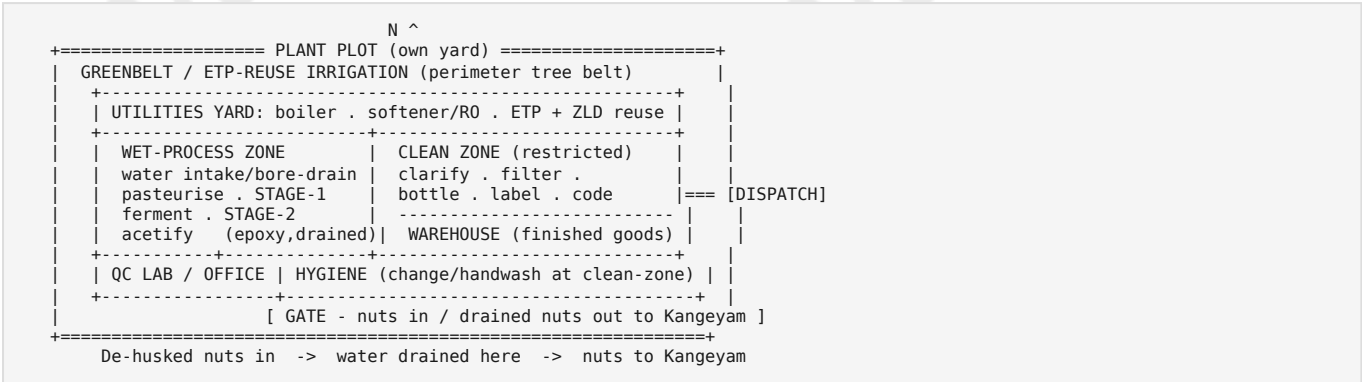
| Risk                                                     | Mitigation                                                                                                                    |
|----------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| Pollution categorisation uncertainty (fermentation)      | Design for the lenient category on the merits (ZLD + ETP, no alcohol sold); <b>written KSPCB confirmation</b> before sanction |
| Copra kernel spoilage if nuts split before humid transit | <b>Bore-and-drain</b> (keep shell intact) rather than full split; quick dispatch to Kangeyam                                  |
| Coconut water perishability                              | Same-day filter + pasteurise; throughput matched to daily nut flow                                                            |
| Low native sugar $\rightarrow$ weak vinegar              | Adjust to $\sim 15^\circ$ Brix with jaggery; QC every batch to $\geq 5\%$ acidity (FSSAI $\geq 3.75\%$ )                      |
| <b>Market development (the real constraint)</b>          | Anchor on bulk food-industry sales; use PMFME/ODOP branding subsidy for the consumer line; e-commerce + export as upside      |
| Packaging cost                                           | Bulk-fill the food-grade line; reserve glass bottling for the premium line                                                    |
| Front-loaded civil (endgame-sized hall)                  | Built once + maxes ESS; ramp needs only tankage                                                                               |
| Regulatory (FSSAI/Legal Metrology)                       | Compliant labels and batch QC from day one; no synthetic-acid fortification                                                   |

## 19. Assumptions & disclaimer

All costs, yields, prices and returns are **planning-grade indicative estimates** (mid-2026 references) for scheme application and bank appraisal. They must be confirmed by: (a) measured daily coconut-water throughput and a lab analysis of the actual water; (b) formal machinery and hall-construction quotations; (c) **written KSPCB categorisation confirmation** and FSSAI

licensing; (d) **written DIC + bank confirmation of ESS + AIF convergence**; and (e) firm packaging and market/offtake commitments before the branded/export revenue is counted. Regulatory categories and subsidy rates are subject to the prevailing KSPCB, FSSAI, DIC and scheme guidelines at the time of application.

## Annexure A — Process & hygienic-flow layout (schematic)



**Notes:** flow is one-directional from the **wet/"dirty" zone** (water recovery, fermentation) to the **clean zone** (bottling, warehouse), with a **hygiene barrier** at the clean-zone entry — the FSSAI-preferred layout. All effluent drains to the **ETP** and is **reused on the greenbelt (ZLD)**. Husk/shell and process residues leave to the biochar unit.

## Annexure B — Term-loan amortisation (Phase-1)

**Assumptions:** term loan ₹66 L; effective rate **6% p.a.** (9% – 3% AIF subvention); tenure **7 years**; **6-month construction moratorium** (interest served, no principal in Year 1); **Kerala ESS ~₹40 L received ~Month 12 and applied as principal prepayment**, leaving ~₹26 L to amortise over Years 2–7 (EAI ≈ ₹5.29 L/yr).

| Year  | Opening (₹L) | Interest @6% (₹L) | Principal repaid (₹L) | ESS prepayment (₹L) | Closing (₹L) |
|-------|--------------|-------------------|-----------------------|---------------------|--------------|
| 1     | 66.00        | 3.96              | 0.00 (moratorium)     | 40.00               | 26.00        |
| 2     | 26.00        | 1.56              | 3.73                  | –                   | 22.27        |
| 3     | 22.27        | 1.34              | 3.95                  | –                   | 18.32        |
| 4     | 18.32        | 1.10              | 4.19                  | –                   | 14.13        |
| 5     | 14.13        | 0.85              | 4.44                  | –                   | 9.69         |
| 6     | 9.69         | 0.58              | 4.71                  | –                   | 4.98         |
| 7     | 4.98         | 0.30              | 4.99                  | –                   | ~0.00        |
| Total |              | ~9.7              | ~26.0                 | 40.0                |              |

The 3% AIF subvention saves ~₹5 L of interest over the loan life; the loan is **collateral-free** under CGTMSE.

## Annexure C — Steady-state cash flow & coverage (Phase-1, base case)

**Assumptions (bulk-dominant base):** output 1.8 lakh L/yr; blended ₹55/L → revenue ₹99 L/yr; operating cost ₹63 L/yr; **EBITDA ≈ ₹36 L/yr**.

| Line                                                                       | ₹ L/yr       |
|----------------------------------------------------------------------------|--------------|
| Revenue (1.8 lakh L @ ₹55)                                                 | 99.0         |
| Operating cost (jaggery, packaging, cultures, wages, power, marketing, QC) | 63.0         |
| <b>EBITDA</b>                                                              | <b>36.0</b>  |
| Debt service (Year 2, EAI on ₹26 L @6%)                                    | 5.29         |
| <b>Net cash before tax/dep.</b>                                            | <b>~30.7</b> |

**Debt-service coverage (Year 2):** EBITDA ₹36 L ÷ debt service ₹5.29 L → **DSCR ≈ 6.8** (comfortably bankable, reflecting the near-free raw material). Branded/export mix lifts EBITDA further (Section 14, upside).

## Annexure D — Machinery quotation-comparison template (fillable, Phase-1)



**How to use:** issue to at least **two local vendors (Coimbatore/Pollachi/Palakkad hub)** and **one specialist food-fermentation vendor**. Ask each to quote a complete **~600 L/day line** with **pre-provisioned tie-ins for Phase-2/3 tankage**, food-grade SS-304 contact parts, guaranteed acidity outcome and delivery terms; carry the total into **Section 11**.

**Table D-1 — Scope & required specification (issue to vendors)**

| #  | Equipment                                                       | Sized for             | Required specification / notes                                     | → Cost item |
|----|-----------------------------------------------------------------|-----------------------|--------------------------------------------------------------------|-------------|
| 1  | Bore/drain & collection station                                 | ~2,000 nuts/day       | Hygienic water recovery + coarse filter + collection tanks         | #3          |
| 2  | Chiller / cold hold                                             | 600 L/day             | Same-day hold of drained water before pasteurising                 | #3          |
| 3  | Steam boiler (small) + plate pasteuriser                        | 600 L/day             | 60–65 °C hold; CIP-compatible                                      | #4          |
| 4  | Alcoholic fermenters (SS-304, jacketed)                         | 600 L/day, expandable | Sized for ~7–14 day cycle; pre-laid tie-ins for Phase-2/3          | #2          |
| 5  | Acetators / surface-acetification vats                          | 600 L/day, expandable | Aerobic; mother-of-vinegar; state cycle time                       | #2          |
| 6  | Clarifier + filtration (plate/cartridge)                        | 600 L/day             | Bright filtration for food-grade line                              | #5          |
| 7  | Bottling line (rinser/filler/capper/labeller/coder) + bulk-fill | 600 L/day             | Glass for premium; drum/can for bulk                               | #5          |
| 8  | ETP + ZLD reuse                                                 | plant effluent        | Anaerobic+aerobic; treated water to greenbelt; <b>no discharge</b> | #6          |
| 9  | QC lab set                                                      | —                     | Titration, pH, alcoholometer, incubator                            | #8          |
| 10 | Erection, commissioning, culture establishment, training        | —                     | Include trial batches to ≥5% acidity                               | #2–#5       |

**Table D-2 — Price & commercial comparison (fill from quotes)**

| Parameter                                                                                      | Vendor A | Vendor B | Vendor C |
|------------------------------------------------------------------------------------------------|----------|----------|----------|
| Make / model offered                                                                           |          |          |          |
| Rated capacity (L/day) & fermentation cycle time                                               |          |          |          |
| Guaranteed acidity outcome (% as acetic acid)                                                  |          |          |          |
| Expandability to Phase-2/3 (tie-in provisions)                                                 |          |          |          |
| Material of construction (contact parts)                                                       |          |          |          |
| Connected electrical load (kW)                                                                 |          |          |          |
| ETP compliance vs KSPCB norms / ZLD                                                            |          |          |          |
| Delivery period (weeks from PO)                                                                |          |          |          |
| Warranty (months) & after-sales distance                                                       |          |          |          |
| Payment terms                                                                                  |          |          |          |
| <b>Total price (incl. GST)</b>                                                                 | ₹_____   | ₹_____   | ₹_____   |
| <i>Target ≈ ₹22 L for fermentation + finishing + bottling (items #2, #4, #5 of Section 11)</i> |          |          |          |

**Evaluation guidance:** weight **guaranteed acidity outcome, ZLD/ETP compliance (for the CTO), food-grade construction, expandability in place, and after-sales proximity** (favouring the Coimbatore/Pollachi/Palakkad hub).